SAY WATT ROBOTICS · FIELD LOG

Week 2.7 — The Schedule Ships, the Float Gets a Flowchart — and a Short Circuit Teaches Us Why Fuses Matter

Sunday, November 1, 2026 · 3:00 – 5:00 PM · 11 students

Eleven of us at 2.7 — Trenton, Isaac, Annabelle, Tyler, Luke, Liam, Rebecca, Oliver, Eli, Frankie, and Rhylan — from 3:00 to 5:00. The headline is a document, and it is a big one. The integrated master schedule — the one calendar that shows what every sub-team owes and when — is finished and uploaded to the team's Google Drive. It first went on the action list at 1.9, back on August 16. Eleven weeks is a long time for one document, and a schedule that is nearly done schedules nothing; now this one is done, just ahead of the full 2027 manual. The rest of the room was busy. Software drew the profiler's brain: a flowchart that walks the float through five steps — start up, descend, hold and log, surface, transmit — with three jobs underneath it: depth control, data collection, and getting the data back to the surface. We also learned PID control, the feedback loop that will hold the float at depth. Mechanical worked on the manipulator's wrist gearing and housing, kept building the full profiler assembly, and modeled stand-offs for the buoyancy engine. The O-ring test bed came off the printer and still doesn't seal, so it goes back to research. And electrical learned a lesson the hard way: the battery's positive and negative leads touched, and we think the short burned out the stepper motor driver. Two new drivers are on order, and protecting connections and fusing them properly is now written into our notes. Marketing is still looking for a new lead.

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STUDENTS AT THE SESSION	WEEKS FROM FIRST ASSIGNMENT TO A FINISHED SCHEDULE	STEPS IN THE PROFILER'S MISSION FLOWCHART	REPLACEMENT STEPPER DRIVERS ON ORDER

WHAT WE LEARNED

- Finished is a feature. A schedule that is ninety percent done for eleven weeks schedules nothing; once it is on the shared drive, every sub-team can plan against it.
- A flowchart turns one big program into jobs people can own. Once the profiler's logic was drawn as depth control, data collection, and comms, the next step was obvious: give each branch a developer.
- PID control is how you hold still in moving water. The controller looks at how far the float is from its target depth, how long it has been off, and how fast that error is changing — then nudges the syringe to close the gap without overshooting.
- Shorts are fast, and fuses are faster. When the battery leads touched, the stepper driver paid for it. A fuse in the right place opens before parts cook — which is why the 2027 float rules require one within 5 cm of the battery's positive terminal.
- A test bed that fails is doing its job. The O-ring test bed still doesn't seal, and finding that on the bench beats finding it four meters down.
- The schematic comes before the board, and consistency is how a drawing gets read. If the same kind of part always looks the same, a judge — or a teammate at 9 p.m. the night before competition — can read the schematic at a glance.

DECISIONS MADE

- The integrated master schedule is finished and lives on the team's Google Drive, where every sub-team can plan against it.
- The profiler's software is organized as a state machine — start up, descend, hold and log, surface, transmit — with depth control, data collection, and communications as separate branches.
- Depth will be held with a PID loop: the depth sensor reports, the controller decides, and the stepper-driven syringe moves.
- Two replacement stepper motor drivers were ordered after a battery short took one out.
- Schematic consistency gets settled before the board is laid out: whether symbols should share a size and shape is now an action item, not an afterthought.

ACTION ITEMS

- Finish the profiler's full assembly and master sketch
  Mechanical group · by 2.8
- Fix the O-ring end cap
  Mechanical group · by 2.8
- Research stand-offs for the buoyancy engine
  Mechanical group · by 2.8
- Decide whether schematic symbols need a consistent size and shape — and how to make them match in KiCad
  Electrical group · by 2.8
- Keep building the profiler PCB
  Electrical group · by 2.8
- Name an owner for each piece of the profiler code: the depth control loop, sensor data collection, and the comms path
  Software group · by 2.8

STUDENT LEARNING OUTCOMES

- Finished the integrated master schedule and published it to the team's Google Drive.
- Drew the profiler's software logic as a state-machine flowchart and learned the principles of PID depth control.
- Modeled the manipulator's wrist gearing and housing in CAD.
- Continued the profiler's full assembly and modeled stand-offs for the buoyancy engine.
- Printed an O-ring test bed and found that the seal needs more work.
- Worked on the profiler's electrical schematic in KiCad and traced a burned-out stepper driver to a battery short.

LOOKING AHEAD

Split the profiler code into owned pieces — depth loop, sensor data, comms — and start writing them against the flowchart. Mechanical finishes the full assembly and goes back at the O-ring seal; electrical settles the schematic symbol question, keeps building the board, and gets the new stepper drivers in when they arrive. The full 2027 MATE competition manual is expected this month; when it lands, the new schedule gets checked against the real deadlines.

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